Education Plan for WATERS Net Incorporated

The WATERS Network Project Office Education Committee

Elizabeth Eschenbach and James Johnson, co-chairs[1]

[1] Full committee membership is listed in Appendix 9.6

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EXECUTIVE SUMMARY

The **WAT**er and **E**nvironmental **R**esearch **S**ystem (WATERS) Network is an infrastructure of environmental field facilities that will establish a national environmental observatory and engineering analysis network to facilitate a comprehensive approach designed to understand large-scale human-stressed environmental systems. In this program, the Network will provide the means to integrate research and education by providing access to professional communities that are focused on many important aspects of environmental quality.

This document (1) identifies the need for the integration of educational activities within WATERS Network and presents the strategic educational goals; (2) examines constructs which describe a transformation of education via the WATERS Network within constituent populations; (3) identifies infrastructure requirements, makes recommendations for assessment and evaluations, suggests approaches for working with other environmental observatories and describes funding strategies; and, (4) presents examples of scenarios for communicating how constructs and diverse populations integrate to transform education via the WATERS Network.

Because of its network structure, the WATERS Network will have many potential impacts on our environment. It is envisioned that it will strengthen the Science, Technology, Engineering and Mathematics (STEM) pipeline, a much needed effort in our society today. It will also transform education at all levels, ranging from improved secondary school curricula to the preparation of next generation scientists and engineering at participating universities.

More specifically, it is anticipated that the WATERS Network will lead to:

1. Modernization of curricula content and pedagogical approaches to reflect critical competencies for the 21st century. This includes trans-disciplinary instructional units and project learning that integrate advanced information and instructional technologies into teaching, learning, and advocacy of water resources issues.

2. Reciprocal relationships between research and education and increased vertical collaboration among all levels within the education community around water quality and quantity issues.

3. Sustained professional development for students, educators and researchers throughout the educational pipeline.

INTRODUCTION

The **WAT**er and **E**nvironmental **R**esearch **S**ystem (WATERS) Network has the potential to significantly change the way students, researchers, citizens, policy makers and industry members learn about environmental problems and solutions regarding water quality, quantity and distribution. Through the use of technology that connects students, faculty, researchers, policy makers and others, WATERS Network can provide learning opportunities and teaching efficiencies that can revolutionize environmental science and engineering education. WATERS Network will provide the means to further integrate research and education by providing access to professional communities that are focused on important aspects of environmental quality. Thus students and others can share and participate in the development of the outputs of these research communities.

This document has been prepared by the WATERS Network Education Committee as part of the WATERS Network Project Office's effort to develop a plan for the education and outreach component of WATERS Network, Incorporated and has been reviewed by members of the hydrologic science community, represented by CUAHSI. In preparing this document, the Education committee reviewed planning documents from the six other WATERS committees as well as existing education plans for other environmental observatory networks, including National Ecological Observatory Network (NEON) and Ocean Research Interactive Observatory Networks (ORION).

The document is organized into four general components. In the first component (Sections 1- 3), the need for the integration of educational activities within WATERS Network is established and the strategic educational goals are presented. In the second, (Sections 4-6), three constructs are examined to describe a transformation of education via WATERS Network within constituent populations. In the third, (Sections 7 -10) infrastructure requirements, recommendations for assessment and evaluation, approaches for working with other environmental observatories and funding strategies are described. Lastly, (Section 12) a set of example scenarios are presented to communicate how the three constructs and diverse populations integrate to transform education via WATERS Network.

1 WATERS NETWORK'S IMPACT ON EDUCATION

Effective educational efforts, both formal and informal (in tandem with outreach to the broader community) are critical for the success of the WATERS Network and are reliant on the high-quality scientific output of the proposed national environmental observatory and engineering analysis network. Educational programs based on state of the art delivery of cutting edge research have the potential to transform environmental science and engineering education at all levels, bringing about broad-based, systemic change that will, in turn, strengthen the Science, Technology, Engineering and Mathematics (STEM) pipeline. The National Research Council notes that "if CLEANER (aka WATERS Network) and the other environmental observatory initiatives fail to transform environmental education and outreach, then they will fail to meet their full potential. The observatory initiative has the potential to make knowledge of the status of environmental resources a basic component of day-to-day life." (NRC, 2006) It is imperative that recognition of the potential substantial impact of this plan on environmental science and

engineering education as well as STEM workforce development will only occur with the provision of adequate, dedicated resources commensurate with the goals set forth.

1.1 Need to Increase the STEM Pipeline

Multiple reports point to the decline in recruitment and retention of students studying STEM as well as to the increased rate of professionals leaving STEM (NSB, 2003; NSB, 2004; GAO, 2005; NAS, 2006). U.S. Government Accountability Office reported to Congress that despite an increase in college enrollments over the past decade, the proportion of students obtaining degrees in STEM fields has fallen. For example, 27% of students received degrees in the STEM fields in 2003-04, compared with 32% in 1994-95 (Rainey, 2006). The Task Force on American Innovation reports that the number of job openings in STEM fields is currently growing at a rate more than five times higher than the number of U.S. students graduating in those fields. As an example, in the area of environmental engineering, the U.S. Bureau of Labor Statistics anticipates that the number of environmental engineering jobs will increase by more than 38% in the next 10 years and Fortune Magazine lists environmental engineering as one of its top 20 careers. However, the number of students studying environmental engineering in the U.S. has not increased to meet this coming demand. Furthermore, the National Science Foundation (NSF) has noted that the lack of an available, scientifically competent workforce looms as a significant limiting factor to the development and deployment of sensor networks required for large environmental observatory systems (Sensors for Environmental Observatories, 2006).

Addressing these disparities is paramount to keeping the U.S. scientifically competitive in the future (NAS, 2006). Many suggest increasing outreach to “previously underutilized groups” at the K-12 level, especially to female and minority students, and increasing the use of mentors to encourage enrollment in those fields (NSF, 2003; Rainey, 2006; NAS, 2006). The WATERS Network has the potential to improve the recruitment and retention of students in STEM by facilitating authentic research experience all along the educational pipeline. Furthermore the Network can deliver practical application-based environmental engineering curricula through multiple effective pedagogies, both actual and virtual. By providing access to high-tech, state of the art, interactive internet-based curricula, the WATERS Network can also provide professional development for K-12 teachers as well as university researchers.

Studies indicate that – for pre-collegiate and collegiate education - one of the most critical factors in student achievement for STEM is the training and expertise of the classroom teacher. The Committee for Economic Development (CED) finds that almost a third of high school math classes are taught by teachers who did not major or minor in mathematics. According to the National Center for Educational Statistics (NCES), in 1999-2000 (the most recent year for nation-wide data), 70% of the U.S.’s largest urban school districts had vacancies in mathematics, 61% had vacancies in biology/life sciences, and 51% had vacancies in the physical sciences. Observations collected by the National Council on Teacher Quality indicate that it is extremely unlikely that every state will meet the new federal requirement that there be a “highly qualified teacher” in every classroom in the nation by the end of academic year 2006. The first recommended action from *Rising Above the Gathering Storm* (NAS, 2006) is to increase America’s talent pool by vastly improving K–12 science and mathematics education via a) 10,000 scholarships a year for new teachers, and b) improved current teacher’s skills via summer

programs, masters of science education programs and Advanced Placement (AP) Science and International Baccalaureate (IB) training programs.

The WATERS Network will play a critical role in developing strategies for filling the nation's classrooms with instructors who are qualified in the physical and life sciences, the application of those disciplines to environmental science and engineering, and modern methods for effective teaching.

1.2 WATERS Network Potential Educational Impacts

The benefits of the WATERS Network to transform education are numerous, ranging from improved secondary school curricula to the preparation of next generation scientists and engineers at participating universities. The WATERS Network can potentially:

- Provide real world data for exploration and demonstration by students from K-12 through graduate and informal audiences.
- Encourage and provide the mechanism for learners to investigate local environmental problems with a global perspective via the integration of research and education.
- Enhance the relevance and quality of instructional materials with a focus on water quality, quantity and its distribution.
- Provide access to online environmental data acquisition instrumentation.
- Provide communication mechanisms to link educators and their students with research scientists.
- Provide a basis for learning about environmental policy through simulations.
- Provide a mechanism for high school and college students to conduct authentic environmental science and engineering research via real-time data and analysis tools.
- Provide structured experiences for graduate students (assistantships, etc) to prepare them for leadership roles in science and engineering.
- Provide professional development for educators on incorporating real-time scientific data, tools, and analytical techniques into classrooms.
- Allow university faculty members access to contemporary research findings to enhance their research and teaching.
- Provide collaborative networks for educators, researchers and students to share and exchange ideas and approaches.
- Create methods for conducting educational research on how learners acquire and interpret information about the environment via the WATERS Network.

2 EDUCATIONAL GOALS

The WATERS Environmental Science and Engineering Plan emphasizes a comprehensive model for examining how the natural world and the "designed" world interact in a systemic fashion for major issues of water quantity, water distribution, and water quality. The plan calls for an overarching view of the interactions among humans, habitats, and hardware, modeled as a set of on-going processes. Central to this focus on system dynamics and societal context is a network of linked databases providing information on natural science, social behaviors, economic interests, and engineered entities within a given geo-physical domain.

This vision of the WATERS Network as a framework of advanced sensors, customizable information technologies, and trans-disciplinary human expertise, culminating in the ability to forecast and communicate environmental conditions, has several major implications for transforming education. First, the emphasis on integration mandates increased attention to teaching systemic thinking and meta-cognitive abilities that underlie strategic and probabilistic reasoning. Second, the reliance upon a cyberinfrastructure calls for improved methods of professional development as the technology of teaching/learning becomes increasingly complex. Third, emerging models of collaboration and communication call for greater attention to concepts that blend both formal and informal learning and teaching venues.

All programs sponsored by or incorporated within the WATERS Network would include an education component. At a minimum, each researcher's project would embed educational outreach activities in his or her monitoring, experimental or computational research. Example activities include training participants (K- graduate students and educators, industry stakeholders, citizens and researchers) on sampling and preservation methods, use of data mining tools to identify leading indicators of potential water quality degradation, or identifying and filling water quality and quantity data gaps. Each "WATERS" proposal would identify the target audience for educational activities and describe how the success of the proposed activities will be assessed. Additionally, the WATERS Network Education Office will continually identify and communicate educational best practices contributed by these sponsored projects.

2.1 Strategic Goals

The Education Committee has identified four strategic goals foundational to mediating change in education:

- Bring together educators, scientists, engineers, administrators, and citizens to form a powerful collaborative that will transform the current state of formal and informal education in environmental engineering and hydrologic science.

- Propagate "best practices" in education that are informed by rigorous cognitive and pedagogical research in order to create a diverse, internationally competitive workforce.

- Enable synergistic interactions among scientists and pre-collegiate/collegiate/graduate educators in setting research agendas and distributing results for the benefit of society.

- Provide broadly accessible, state-of-the-art information bases and shared research and education tools.

From these goals, a range of educational reform objectives evolve – covering delivery of instruction, learning outcomes, teacher/instructor training, professional development, and societal impacts.

2.2 Constructs For Transforming Education

The Education Committee has targeted three major constructs for driving change, which are given more complete development in Sections 4-6. The committee recommends that activities sponsored by the WATERS Network address as many of these functional areas as possible.

- Modernization of curricula content and pedagogical approaches to reflect critical competencies for the 21st century. These include trans-disciplinary instructional units and project learning that integrate advanced information and instructional technologies into teaching, learning, and advocacy of water resources issues.

- Reciprocal relationship between research and education and increased vertical collaboration among all levels within the education community around water quality and quantity issues.

- Sustained professional development for all participants.

Additionally, the Education Committee has identified five critical constituencies to serve as agents within the above areas of focus. “Cradle to grave” education is addressed by these five populations: K-12, Undergraduate, Graduate, Industry and Citizens.

2.3 Guiding Principles for Promoting Inclusion and Diversity

In all the constructs for transformation and for all constituencies described above, the WATERS Network will be committed to the integration of processes that promote inclusion and diversity in its educational and scientific goals using the following guiding principles:

- Embrace our responsibility to create an inclusive and respectful climate that enhances the scientific and educational experience of researchers, staff, educators and students involved with the WATERS Network.

- Acknowledge the critical role of the WATERS scientific community, in collaboration with their educational partners, in developing and maintaining an environment that invites the participation of a diverse constituency.

- Recognize that an inclusive climate is rarely generated or maintained in the absence of supporting policy, structure, and oversight, and commit to building systems supportive of inclusion and diversity into all strategic plans, both scientific and educational.

The list in Appendix 9.2 that contains a list of practices that will promote inclusion and diversity should be carefully considered by the Organization Committee.

3 MODERNIZING CURRICULA AND PEDAGOGICAL APPROACHES

The WATERS Network seeks to develop education and outreach resources that will advance the understanding and appreciation of environmental science and engineering by students at all levels. Students will learn about impacts of human activities and natural perturbations on the quantity, distribution, and quality of water. In order to achieve this, the WATERS Network will (1) provide accessible data on water quality, water quantity, and human drivers of environmental impacts; (2) create the ability to investigate and analyze environmental conditions from this data; and, (3) create technology-based applications and activities that will promote enthusiasm and ownership of learning.

The access to environmental data through the network can meet the goals of 21st century education in delivering maximum content in ways that create knowledgeable and self motivated learners who are equipped to find and understand new knowledge and apply it to their needs. Students of the 21st century have been referred to as "digital natives" (Prensky, 2001). Learning that does not use technology, quickly loses its relevance to the digital native. The WATERS Network will provide a technologically advanced avenue for environmental education that will motivate students to become active learners by teaching with technology using appropriate curricula and pedagogy.

3.1 Processes for Modernizing Curricula and Pedagogy

3.1.1 Curriculum Development

Real-time, comprehensive data that will become widely available because of the WATERS Network provides new opportunities for truly unique curriculum development at all levels of formal and informal education. These curricula would be aligned with primary and secondary educational standards. The Network will allow undergraduate and graduate educators to be involved in the development of new curricula to help achieve existing science and engineering educational objectives (e.g., ABET outcomes for engineering programs).

Relevant curriculum would be developed for governmental, non-governmental, and industrial users of the WATERS Network. Within the general population, the WATERS Network will establish informal education opportunities that employ an iterative approach allowing participants to learn about environmental engineering problems while contributing to the Network database via citizen science.

The WATERS Network would engage science and technology museums to mediate between social needs and the practices, techniques, and culture of large scientific projects (Chittenden et al., 2004). To support curriculum development, the Network will provide and update the necessary training information for users, at all levels, to easily access and use the WATERS facilities. All of these curricula development efforts will achieve the goals of the Network concerning the importance of understanding and designing within the water cycle and the consequences of poor practices.

3.1.2 Pedagogical Enhancement

Pedagogical strategies that promote active learning would be developed using the access to real-world and real-time environmental data available through the WATERS Network. Curricular standards across the nation tend to be fairly uniform, but pedagogy varies widely with the type of institution and teacher preferences. While both curriculum and pedagogy are equally important, if the pedagogy is not well received by the students, the content is often not assimilated. Pedagogy then becomes the critical link between content and true learning.

Students would access and synthesize real environmental data from the WATERS national network of environmental sensors. The WATERS Network will have an interface allowing users to easily access and query data at multiple levels. The WATERS database would allow the examination and comparison of state and/or regional environmental parameters. The availability of comparable, geographically disaggregated data will be valuable for primary and secondary students, undergraduate and graduate instructors, and researchers in conducting their teaching and research. The Network would link regional, and national environmental data to state water quality standards through their Environmental Protection Agency (EPA) and Departments of Natural Resources (DNR), thus increasing the connection between WATERS data and real-world practice.

The WATERS Network will provide students sample environmental problems and issues for research projects. Tutorials and animations will guide students through the data in their investigations. Depending on student age, ability level, and the depth of their investigation, they will be encouraged to collect samples in their local area and connect the findings to that produced from the regional data obtained from the WATERS Network.

3.1.3 Educational Research to Identify Pedagogical Best Practices

Educational researchers could use the Network to provide information about how people learn and what pedagogical approaches work best. The Network could also be used to assess optimal approaches for learning of different content and by different populations. Best practices could then be broadly disseminated using the Network.

3.2 Outcomes from Modernizing Curricula and Pedagogies

Through active engagement with environmental data, students would appreciate ways in which human activities impact the quantity, distribution, and quality of water. Students would focus on water resources issues that affect them directly and propose solutions using WATERS data, increasing the relevance of their education and research. In developing new educational tools, data from the Network would be integrated for use by students, educators, and industrial and community stakeholders. For example, at the K-12 level, course projects based on water quality monitoring assessment, and management can be developed using WATERS Network summary reports. The availability of summary data at regular intervals can provide the opportunity to use longitudinal data as a context-based teaching tool.

As the Network is used by educators, a community with expertise in applying cyberinfrastructure-based information to environmental science and engineering education would

develop. This expert community could provide “textbook” examples that were found to be particularly effective components of curriculum or pedagogical practices.

For undergraduate and graduate students, real-time and archived data from the WATERS Network would be available to examine the impacts of human activities on regional water systems. The Network will facilitate creation of environmental investigations for science and engineering courses along with field- and laboratory-based activities. The WATERS Network Project Office will offer summer short courses and workshops at each observatory for students from across the country. WATERS data will facilitate intentional, connected teaching and learning, including hands on, cooperative, context-based, practical application driven opportunities in order to increase the retention of undergraduate STEM students.

WATERS data will be used in educational research, evaluating the impact of problem-based learning, collaborative learning, and other active learning techniques on student learning. Professional curriculum developers will create contextual and problem-based learning educational modules for use by educators which can serve as models for further development of educational activities. Educational research can be used to identify pedagogical approaches that are effective with underrepresented groups or that help recruit and retain more students in STEM. This information will be presented at research conferences in order to integrate the Network into university classes and research.

4 INTEGRATION OF RESEARCH AND EDUCATION

The objectives of integrating research and education both horizontally and vertically are to (a) accelerate the exchange of information and ideas among educators, students, and researchers; (b) promote the development of relationships among K-12, college, and university educators that will be mutually beneficial; and, (c) develop methods for educational research that enable continuous improvement of environmental education practices. The increased interactions among educators at various levels will also advance the broader goals of widening the pipeline of students from K-12 education into undergraduate and graduate education in science, technology, engineering, and mathematics (STEM) and improving the professional development of K-12 science educators. Further, the collaborations that can be developed among university researchers, K-12 educators, and community stakeholders will create a conduit for new populations to have input into the research agenda. An unprecedented opportunity exists for educational research at all levels in order to understand how the Network can be efficiently used to create effective technology-based learning environments for all participants.

4.1 Processes for Achieving Horizontal and Vertical Integration

4.1.1 Forums for Interaction

Conferences on scientific research and education are natural forums for increasing interactions among students and educators from multiple levels. Special sessions can be organized at these meetings that focus on the presentation of research performed by teams of K-12 through post-graduate researchers. Conference symposia that promote vertically integrated research and education can be organized for well-established meetings (e.g., AGU, AEESP, ASCE, ASEE, WEF, ACS) as well as at special workshops and conferences organized by the WATERS Net Inc. Project Office.

Forums for vertical collaboration can also be performed with smaller and more focused groups, including (or modeled after) summer programs such as Research Experiences for Teachers (RET) and academic year collaborations like Graduate Fellows in K-12 Education (GK-12). For the earliest educational levels (K-4), educational components will offer hands-on activities led by teachers or students who have WATERS Network research experiences from RET, GK-12, and related programs. To broaden the participation of traditionally underrepresented groups, many of these forums should be hosted at women's colleges, historically black, Hispanic-serving, and tribal colleges and universities. Graduates of these institutions successfully complete graduate degrees at a significantly higher rate than their counterparts who attend predominantly majority-based institutions.

4.1.2 Developing Network Tools that Enable Reciprocal Relationships

The WATERS Network can provide channels of communication among K-12 schools, colleges and universities, research institutions, government agencies, professional societies, and the general public. The Network will allow students to become active participants in collecting data and informing the research agenda. Mechanisms will be developed that allow students to identify local and regional water issues and report this information to WATERS researchers.

Methods for encouraging feedback from educators, students, and other stakeholders to the researchers about the usefulness of the data, the appropriate format of the data, and suggestions for improving access to data would be developed so that these end-users are active members of the research process and not simply passive recipients of information. The feedback will be particularly helpful for educational research that seeks to evaluate optimal methods for teaching environmental science and engineering.

Inter-university network collaborations can facilitate the incorporation of field research data into undergraduate and graduate curriculum. For example, university faculty will develop research methods courses that allow students to conduct field research from their home institutions and then compare and contrast their data with the data in the WATERS Network. In order to encourage real-time data analysis, the WATERS Net Inc. Project Office could create a national student investigation competition to increase use and awareness of the Network, and potentially attract and recruit more students to pursue careers in science and engineering.

4.1.3 Establishing Multi-level Collaboration as Core Project Components

Effective implementation of vertical collaborations should be included as core components of all WATERS Network projects and not just as accessories to projects performed at only one level of the educational system. Requests for proposals and review of proposals would encourage the involvement of K-12 educators and other non-university stakeholders in both the education and research components of proposed projects. WATERS Education staff would assist in the development of proposals' education and outreach components. Such a requirement would expand upon the benefits already gained by the NSF's requirement that all proposals address and be evaluated on both intellectual merits and broader impacts criteria. The initiation of collaborative relationships among stakeholders (e.g., connecting university researchers with K-12 educators) will benefit from a database of stakeholders that can be developed and maintained by the Network.

4.2 Outcomes of Integrating Research and Education

Long-term relationships will be developed between environmental science and engineering educators and researchers at all levels of the educational system. Educators will incorporate state-of-the-art research results on the impact of human activities on water resources and water quality into their teaching.

Through improved access to research results and relationships with scientists and engineers, students at all levels will understand fundamental principles of water resources and water quality as well as the role of environmental scientists and engineers in addressing these issues.

Researchers will play an increased role in environmental science and education through active collaboration with professional educators, and researchers will more effectively communicate the content and significance of their work to a broad range of audiences.

5 SUSTAINABLE PROFESSIONAL DEVELOPMENT

The WATERS Network will promote continued professional development for students, educators, and researchers throughout the educational pipeline. Realizing the full potential of the network will require that users receive appropriate training on the use of Network tools. Such training is an essential prerequisite or co-requisite to transforming education through curriculum, pedagogy and education-research integration described in the preceding two sections. Support will be provided for integrated models of learning and teaching across all stakeholder populations including our scientific, corporate, and industry partners.

5.1 Processes for Professional Development

Below are several processes intended to mainstream professional development into all Network core activities for all stakeholder populations.

5.1.1 Workshops and Training

The professional development of educators and researchers would be promoted by workshops and training sessions that:

- Provide instruction on the use of technology in research and education to all users of WATERS Network technology. In addition, Principal Investigators (PIs) and WATERS Network funded research staff would visit college campuses on a regular basis to introduce students in graduate and undergraduate science and engineering courses to real-world applications of theoretical research models.

- Provide scientific expertise to faculty in higher education through professional development workshops on current research methodologies, cyberinfrastructure, sensor technology, data streaming, and predictive models enabled by WATERS research. These workshops will also serve as a point of dissemination for educational/curricular resources that can be used to enhance pedagogy, especially in graduate and undergraduate courses in science and engineering.

- Provide information and support to PIs for documenting and disseminating education and outreach activities so these efforts can be included as part of the PIs' retention and promotion efforts.

- Provide regularly scheduled opportunities for the WATERS scientific community to interact with education and outreach professionals who will advise them on effective teaching methods for science and non-science populations, including specific instruction on appropriate strategies for creating and maintaining an inclusive educational environment. These sessions will be mandatory for WATERS Network-funded researchers.

5.1.2 Summer Research Opportunities for Educators

The WATERS Network would team with universities to offer funded continuing education credit to K-12 educators participating in summer research opportunities at WATERS facilities as recommended in *Rising Above the Gathering Storm* (NAS, 2006). Educators would learn about and participate in current environmental research methodologies and state-of-the-art monitoring technologies that emphasize the critical, global role of water research. Participants would then develop environmental science and engineering modules for use in their K-12 classrooms.

5.1.3 Presentations by Undergraduate Students

The network would provide opportunity, technical support, and funding for undergraduate students involved in WATERS Network research to develop professional presentations for conferences and professional meetings. Students who receive funding from the WATERS Network Project Office would also develop age-appropriate presentations for local high school students as part of an aligned K-12 outreach project.

5.1.4 Communication with Stakeholders and Community at Large

To reach a broad audience for professional development WATERS Network Project Office would:

- Host an annual science and policy forum for municipal, state and federal legislators to develop and maintain lines of communication with policy-makers, educate them on current research trends, encourage the use of Network resources (data, technology, expertise), and inform policy decisions.

- Use mass media to routinely update the public on progress being made on environmental issues using outputs from the Network.

5.2 Outcomes from Professional Development

Professional development opportunities provided at the WATERS observatories would produce researchers, educators, and workers trained in environmental monitoring and analysis techniques. As technology continues to change and improve, the Network would provide training to allow participants to upgrade and learn new skills.

Each year that the WATERS Network observatories offers summer research opportunities for K-12 educators, a new cohort of teachers with upgraded research skills, better understanding of current environmental issues, knowledge of and access to resources available through WATERS Network, would return to the classroom. Each teacher's newfound knowledge and enthusiasm would serve as a catalyst for increasing student interest and retention in science and math classes throughout pre-collegiate education. Additionally, modules developed by educators would be pilot tested in the originator's classroom, further developed using the expertise of WATERS Network education professionals, cataloged, housed in a WATERS Network web-based resource library as well as other well-used digital libraries, and made available for wide-spread dissemination. Over a few years, this program will develop a significant cadre of teachers who will in turn reach a significant number of K-12 students.

Through ongoing professional development, a community of graduates with experience in WATERS tools will develop. These graduates will carry their knowledge of these tools and capabilities with them into their professional careers, becoming spokespersons for the Network. In addition, these undergraduate students will serve as outstanding role-models for students who attend their outreach presentations in a K-12 school or community forum.

The public visibility generated by professional development outreach will increase general knowledge of the WATERS Network, enhance its public credibility, and provide a forum for influencing environmental legislation, policy, and funding

6 ORGANIZATION AND FUNDING OF EDUCATION AND OUTREACH PROGRAMS

6.1 Physical and Cyber Infrastructure Requirements

At the national office, and at each observatory, there would be at least one facility for Education, Outreach & Training (E&O) activities. This facility would have the technology and space that would facilitate E&O activities such as computers, projectors, video link, etc, similar to the NCSA ACCESS Facility. The facility would be used to train people on the use of WATERS models, communication/collaboration software and hardware, such as sensors as well as provide a venue for education and outreach activities.

At each observatory, K-12 partners and citizen science organizations would be provided with sensor systems to enhance the core and research data collection. The equipment required for these groups is under consideration.

Appropriate infrastructure is required within the cybercollaboratory that will allow the access to data, analysis tools and communication tools to the all participating communities. In their WATERS test-bed project and in previous test-bed work, Piasecki et al. (2006), have already noted the importance of using a parallel (rather than sequential) approach to developing the necessary cyberinfrastructure required for research and education. They strongly recommend that the design of education and outreach efforts and associated tools be designed in parallel with any tools required for the research effort.

6.2 Organizational Structure and Interaction Models

Based on the draft organizational model proposed by the WATERS Network Organization committee, what follows is an imbedded model of a structure that supports and promotes interaction between Education and Outreach and other components of WATERS Network as well as facilitating interaction across different levels of the education and outreach hierarchy.

The Director of Education & Outreach oversees one of the four peer units reporting directly to the CEO Executive Director of WATERS Net Inc. 1) the CFO of Finance; 2) the COO of Operations; 3) the Research Liaison and 4) Interdisciplinary Research Coordinator (Figure 1). Frequent interaction would occur between the Director of Education & Outreach, the CEO Executive Director, the Standing Committee for Education and Outreach and the Research Liaison, the Integrative Research Coordinator. Communication between the Director of Education and Outreach and other executive units would be routine, but not necessarily as frequent.

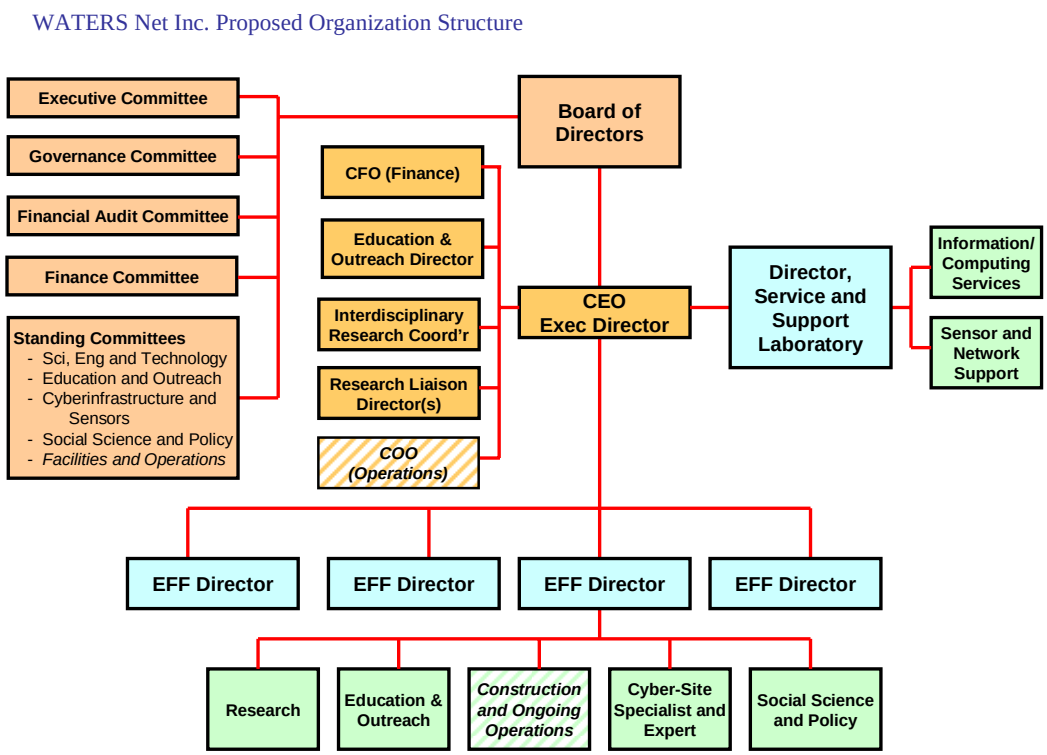

Figure 1. Proposed Organizational Model to Promote Effective Integration of Education & Outreach

Figure 2 outlines a structure for the Office of Education and Outreach that combines necessary hierarchy with an internal organizational structure that seeks to promote collaboration and communication between units. Personnel would be required for managing formal and informal education efforts as well as partnerships. This model also identifies units outside of the Office of Education & Outreach that will be critical to the success of initiatives emanating from that office, such as the Integrative Research Coordinator, Experimental and Field Facility (EFF) Directors and EFF Education and Outreach Directors.

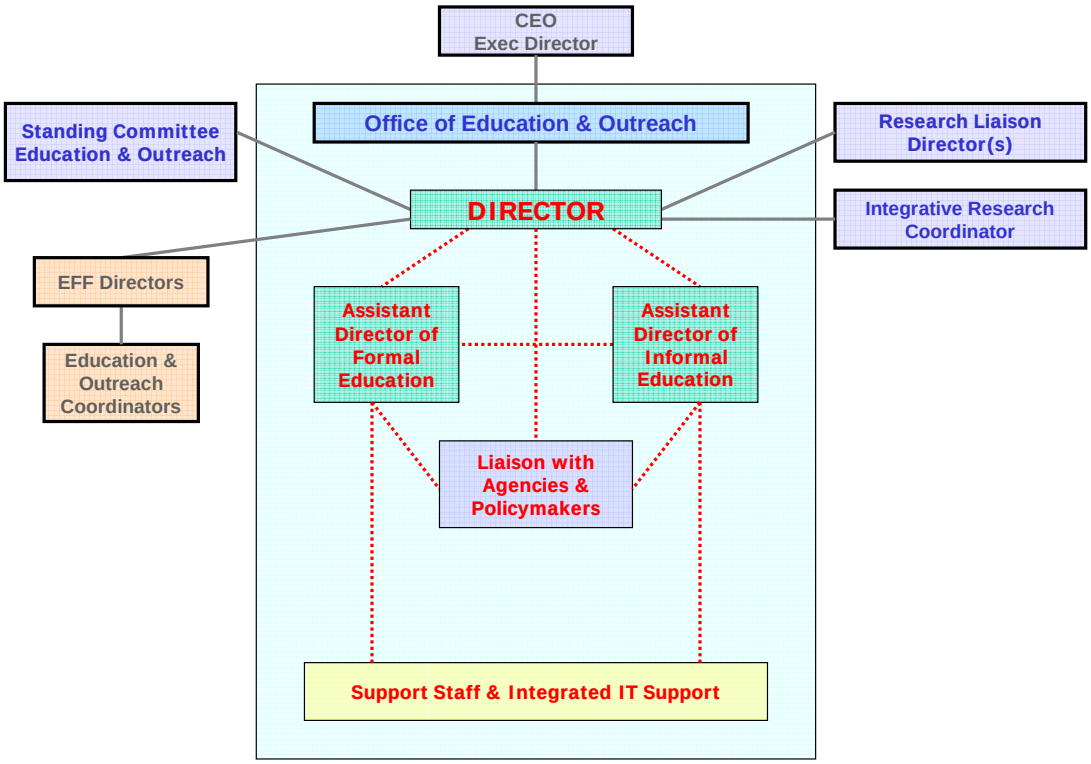

Figure 2. Proposed Structure for the Office of Education & Outreach and its Relationship to Other Units

Figures 3 and 4 depict a structure that supports desired routes of communication, both vertical and horizontal, between units seen as particularly important to the efficient and effective development, implementation and evaluation of programs involving education and outreach. Structures that support an increase in the frequency and quality of interaction between scientists and educators will also increase an early exchange of ideas promoting genuine integration of educational processes into environmental science and engineering research. An additional benefit garnered from building an interactive community of scientists and educators may be a decrease in the time between the acquisition of scientific knowledge and its translation and transfer into the public domain.

Figure 3 identifies routes of direct interaction that need to be developed and maintained by the Office of Education & Outreach and its Director. Figure 4 identifies routes of direct interaction that need to be available to the EFF Education & Outreach Directors.

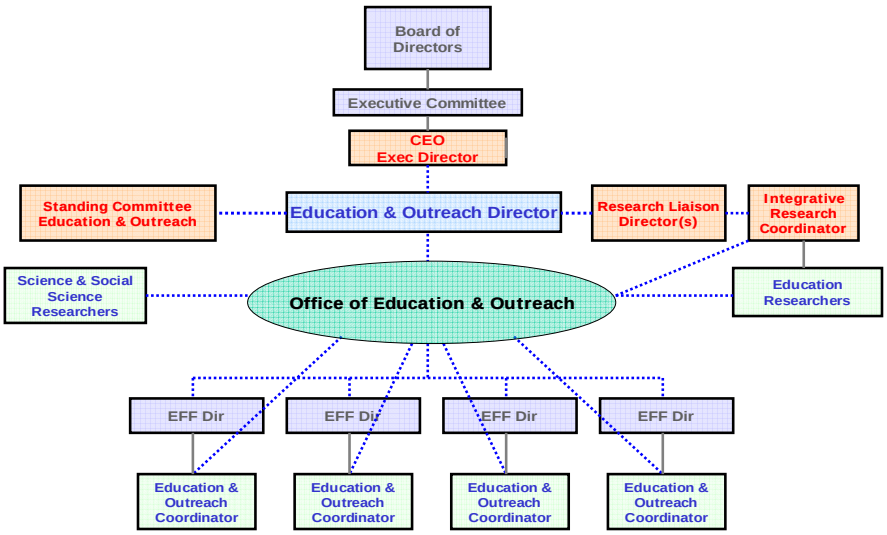

Figure 3. Routes of Direct Interaction Maintained by the Office of Education & Outreach

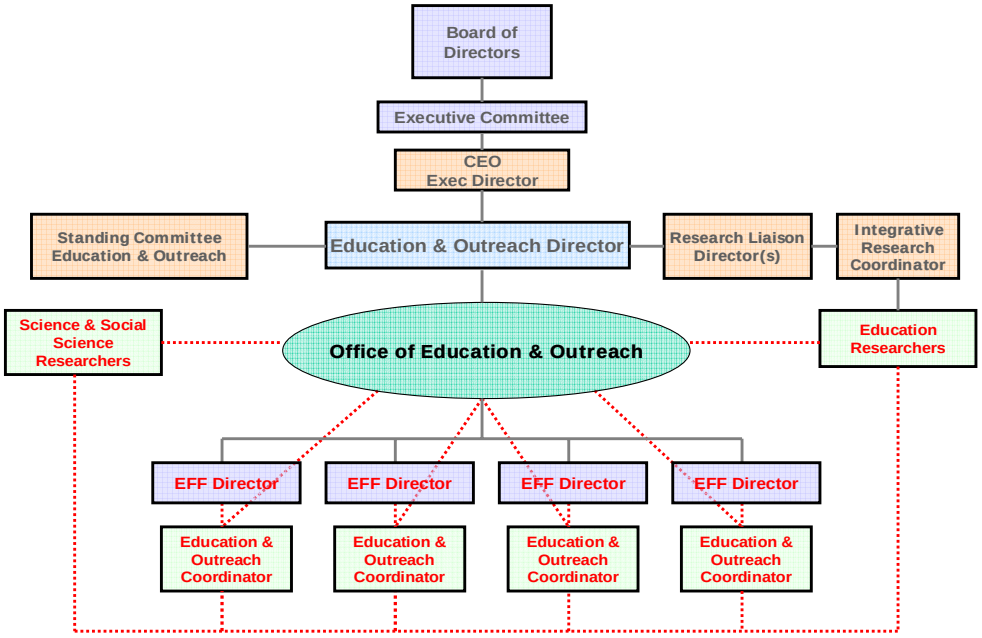

Figure 4. Routes of Direct Interaction Available to EFF Education & Outreach Directors

6.3 Partnerships and Coordinating with Other EOSs

The WATERS Network would lead new partnerships to build new, unique, state-of-the-art educational programs as well as join existing excellent programs to transform environmental education within our schools, universities, professional organizations and communities. Organizations that share education and training concerns with the WATERS Network would be

identified at both the local and national levels. This list includes existing environmental observatory systems, federal, state, and local government agencies, universities and other nonprofit organizations, and existing digital libraries and educational online programs that all may be conducting research or developing or disseminating educational material for cradle to grave education. Appendix 9.3 lists sample organizations for partnerships.

Forming strong liaisons with other environmental sensor networks (e.g. NEON, ORION, those managed by USEPA, USGS, NOAA, etc.) would allow the sharing and exchange data as well as build upon these organizations' successful education and outreach efforts in the environmental education area to school children and their teachers, college students, graduate students, postdoctoral researchers, researchers, industry members, the general citizenry and decision makers. Such partnerships would lead to the development and dissemination of multi-institutional curricular and teaching approaches that use the resources of the WATERS Network and other modern research tools. The National Research Council (2006) recommended that NSF form a parent organization such as "Environmental Observatory Networks" to coordinate the shared activities of the different NSF-funded environmental observatory programs and reduce redundancy. In January 2007, the NSF sponsored a meeting for social scientists to consider how collaborations could occur across the EOs. We recommend that a similar meeting occur to facilitate similar collaborations across the EOs in the areas of Education and Outreach.

At least one staff position at the national office (Figure 2) would be solely responsible for organizing and linking related resources (e.g. observatories, tools, funding, databases, etc.). The WATERS Network liaison staff would continually seek out collaborative opportunities across federal agencies and industry stakeholders for education opportunities as well as garner joint agency support to provide shared data needs for industry. This staff person would develop and maintain relationships with institutions that can significantly increase the number of under-represented minorities in the science and engineering pipeline such historically black colleges and universities, women's colleges and universities and predominately Native American institutions. The liaison will also be responsible for facilitating the formation of mentoring relationships between WATERS Network researchers and students. This staff would identify and network with existing funding programs that can be used in undergraduate and graduate education and research, and thus the influence of the WATERS Network would be leveraged. This staff would identify existing programs and resources relevant to industry stakeholders and determine how WATERS Network can support these activities and possibly integrate the WATERS Network into existing required continuing education activities.

6.4 Funding Models

Funding would be realized by two components: (1) directly through WATERS Network Inc; and, (2) leveraging with other organizations. Both are needed for a successful initiative, and they must be propagated through the entire organization of the Network, as described below.

Direct project funding would maintain the core educational functions of the WATERS Network that are necessary to maintain the interaction of the EFFs with stakeholders. The Network will directly support major educational initiatives that further its mission, thus ensuring its continued relevancy. Targeted funding would be sought to support efforts to create alliances with institutions that have high potential to increase and diversify the STEM pipeline such as

historically black colleges and universities, women's colleges and universities and predominately Native American institutions. In addition, research fellowships can be used to support research related to environmental justice.

The WATERS Network would facilitate direct funding for educational and outreach projects and research that spotlights its education goals. Such projects may include, but are not limited to, program and curriculum development, supplies to support new or existing educational activities, workshops, and research on the best way to engage students and the public in crucial Network issues. These projects can be facilitated by either the Education & Outreach Director if multiple observatories are involved or by the Education and Outreach Coordinator if just one observatory is involved.

Government, non-governmental organizations, and industry all may have an interest in directly or indirectly funding WATERS Network education outreach activities as part of their own outreach programs. Organizations that benefit from WATERS infrastructure would contribute to the education resources that enabled the benefit. The return to these stakeholders is great, compared to not using WATERS Network resources. As a result, all involved organizations effectively leverage funds. The nature of these partnerships, including fees and terms between the WATERS Network and each category of stakeholder, must be negotiated based on the category of the user and frequency of use of the Network. Pursuing large federal and foundation funding that involves multiple observatories and Regional external funding for programming specific to a single observatory will also be the responsibility of the Education & Outreach Director. Smaller regional opportunities will be coordinated as appropriate.

Specific guidance documents and customized workshops would be available to industrial users for a competitive fee. These high-quality educational materials would provide the background to enable industrial stakeholders to efficiently tap into the WATERS resources. These documents would be produced at the headquarters with assistance from appropriate EFF E&O Coordinators and other education researchers. Appendix 9.3 is a draft list of possible funding sources.

7 ASSESSMENT AND EVALUATION

The assessment process will target competencies that reflect the overarching goals of the WATERS Network. Appropriate assessment will allow the Education & Outreach Director (and the standing board committee) for the Network to determine the impact of educational programs and environmental projects as well as evaluate program management of the various components of the Network.

7.1 Guiding Principles

Assessment is the process of documenting – in definable terms – change in knowledge, skills, and/or performance in a given domain. Any large-scale effort with the stated mission of "transforming the formal and informal educational and communication processes" surrounding environmental issues must incorporate a robust assessment plan. All elements of assessment should be informed by a set of guiding principles:

- Use assessment to support improved professional development programs and continuous quality improvement for classroom teachers. Process reinvention in education must

include ways to sustain the constant renewal of competencies necessary for the modern-day educator, including best practices for instructional design and understanding student learning styles.

- Use assessment results in a responsible and ethical fashion to support high-level educational reform movements. Include stakeholders at all levels in the distribution of results.

- Ensure equity through ethnic, racial, economic, and gender fairness.

- Sponsor a wide variety of assessment types – beyond the traditional quantitative formats – to include approaches such as performance-based measurements, portfolios, and student-designed assessment.

7.2 Program-Level Evaluation

As currently conceptualized, the education function of the WATERS Network is embedded within several proposed institutional sites which will coordinate scientific collaboration and the dissemination of results. Therefore, assessment and evaluation must be accomplished at multiple levels within the totality of the program. At a minimum, comprehensive program evaluation must take place both at the Executive Director's level and at the EFF level as described in the WATERS Network Organization document.

The Education Committee proposes that both the Director's Office and the EFFs conduct program evaluation in accord with the principles and methodologies advocated by the National Science Foundation and others (NSF, 2006; Academic Quality Improvement Program, 2006; Baldridge National Quality Program, 2006 and Green and Tineke, 2002)

7.3 Project-Level Assessment

All projects supported by and contributing to the WATERS Network's educational goals would subscribe to modern assessment practices and would monitor (1) process; (2) outcomes; and, (3) impacts of the activities (North Central Regional Educational Laboratory, 2006). The following types of assessment might be used:

- Automated tracking of usage patterns for informational websites and web-delivered instructional software. Analysis of target group changes in performance and attitudes through surveys, achievement testing, and collection of outcome artifacts.

- Embedded measurements and other forms of distributed assessment that contribute to teaching critical thinking and other skills necessary for the 21st Century workplace (e.g. portfolios, longitudinal tracking, student-designed assessment and self-reflection).

- Assessment that encourages standards-based teaching and fosters sustained professional development (constructing engaging classroom activities, writing rubrics/scoring devices that reflect learning objectives, and sharing results among other teachers)

- Systemic testing that encourages students to develop high-level cognitive capacities (e.g., Bloom's Taxonomy and the Perry Scheme) by seeing connections among content areas and real-life applications. Alternative assessment forms that require students to produce a response rather than select from a list of possible responses.

- Multidimensional assessment programs that capture students' growth and fluency with knowledge, ability to self-manage learning tasks, and perform multi-staged problem solving. Such assessment moves from the de-contextualized and atomistic testing of discrete skills to a more orchestrated use of concepts within a contextualized process.

8 EXAMPLE SCENARIOS FOR TRANSFORMING EDUCATION USING WATERS NETWORK

8.1 High School Science Project

Ms. Eckstein teaches science and incorporates real-time data into the course. Using real data in the classroom provides an aspect of authenticity to classroom projects, teaches students how to interpret and manipulate data, while meeting the required teaching standards for science and technology in the state. For assignments on water resources (e.g., water supply and flooding), water quality, and water treatment, her students use the WATERS Network sensors on and nearby campus as well as data available on the Network to explore environmental science concepts. Activities include examining the relationships between water quantity and water quality in different regions, looking at the impacts of water quality on aquatic ecosystems, and evaluating the fate and transport of pharmaceuticals, personal care products, pesticides and herbicides in the aquatic environment.

For a course project, students work in groups to examine specific impacts of human activities on different regional water systems (each group is assigned a different region). In addition to evaluating existing information, each group is charged with the task of developing a hypothesis on human-water system interactions and using the WATERS Network to test the hypothesis. The students' project reports are made available to the WATERS Network cybercollaboratory. Professor Cruz, who also uses the WATERS Network, is conducting research in an area of one of the group's hypotheses. He reads the student report and contacts Ms. Eckstein. They arrange for a videoconference to discuss the hypothesis with the class. As a group, they realize that additional field sites would be necessary to truly test the hypothesis using the WATERS Network. Ms. Eckstein collaborates with Professor Cruz and the local WATERS EFF to write a joint proposal to develop the additional field sites and educational materials.

8.2 Graduate Student Researcher

Ms. Thompson is a Ph.D. student studying the fate of pharmaceutical chemicals in wastewater treatment plants in bench-scale laboratory research. As she nears the completion of her dissertation, she wants to know more about the concentrations and fate of the compounds she has been studying once they are released to the environment. Her advisor suggests using data from the WATERS Network. She finds interesting information, but none related to pharmaceutical compounds. Because Ms. Thompson seeks to complement her expertise in lab-scale experimentation with field-scale research, she signs up for a three week summer program on field sampling and data analysis. The programs are organized by the WATERS Network Project

Office and held each year at one of the EFFs. Ms. Thompson learns several new skills at the workshop. In addition, she has the opportunity to discuss her interests in pharmaceutical compounds with Dr. Carpenter, the director of the EFF. They use their complementary expertise in field monitoring and lab-scale analytical chemistry to develop a plan for monitoring selected compounds at the facility. Ms. Thompson integrates this work into her Ph.D. dissertation and then works as a post-doctoral researcher at the EFF further developing the sampling and analysis protocols.

While continuing her research at the observatory, now Dr. Thompson is joined by Ms. Nguyen, a graduate student in civil engineering and Mr. Thilges, a high school earth science teacher. Ms. Nguyen and Mr. Thilges work as interns with Dr. Thompson over the summer learning sampling techniques and data analysis. Ms. Nguyen creates a poster presentation and paper on her research which she presents at an environmental engineering conference as a part of her master's studies. Mr. Thilges collects data which he incorporates into his classroom study of the effects of human activities on water quality. He also earns graduate credit in environmental engineering from a University associated with the EFF which he applies toward his continuing education requirements.

8.3 Educational Research

Dr. Akoev is a professor of science education at a university conducting research on the impact of analysis of real time data on middle school science students' ability in STEM. She designs a study to examine the impact of the real-time data lessons on assessment scores in math and science and on student's math and science understanding, with a focus on underrepresented students in STEM. She works with a group of middle school teachers to develop lessons that incorporate WATERS data into the classroom study of data analysis and water quality. The lessons focus on statistical sampling and graphical analysis integrated with learning common water quality parameters. Students are assessed and test scores are examined to determine the effect of the lessons on math and science learning. In addition, students' attitudes toward STEM careers are assessed before and after the lesson. These results are disseminated via the WATERS community and elsewhere to improve the accessibility of STEM to all students.

8.4 Community Watershed Association

Ms. Edmonds has been leading a volunteer water quality monitoring group for her watershed association for the last ten years and hears about the WATERS Network at a conference. She contacts Dr. Carpenter, the director of the local WATERS Network EFF, to seek possible collaborations. Dr. Carpenter suggests several options. First, he invites Ms. Edmonds and other members of her group to a summer workshop on field sampling methods and data validation. Dr. Carpenter is particularly concerned about data validation because he wants to protect the quality of data available through the WATERS Network. Dr. Carpenter also suggests that the group may wish to "adopt" a local sensor, monitoring the data produced by the sensor for accuracy and going into the field as necessary to maintain the sensor and correct any problems that arise. The volunteers in Ms. Edmonds' organization engage with both opportunities and as a result become more enthusiastic about their project, knowing that they are providing valuable information and services to the larger network.

9 APPENDICES

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9.2 Strategies that Support Inclusion and Diversity

The following strategies will support inclusion and diversity in WATERS educational components:

- Filter decisions about the geographic placement of WATERS Experimental Facilities and field stations through a lens of current, context-based public health, environmental health and environmental justice concerns. This analysis will provide an avenue to facilitate greater interest in the activities happening at that data collection point and will increase investment by local and regional community and educational stakeholders in those activities.

- Develop inclusive language to be included in all materials used in the hiring process as there is a documented effect on who applies for a position. Set and follow guidelines for what constitutes a 'diverse candidate pool'.

- Require all persons interviewing for positions funded through the WATERS Network to demonstrate a commitment to inclusion and diversity by citing concrete examples of how they foster an inclusive environment within their sphere of influence, no matter how large or small. Applicants that cannot point to specific examples are less likely to be committed to inclusion and diversity.

- Include at least one performance standard that benchmarks an employee's contribution to maintaining an inclusive workplace climate as part of every performance evaluation. Making everyone responsible for their immediate environment contributes to a shared investment in maintaining the overall environment.

- Provide opportunities for members of the WATERS scientific community to interact with equity professionals/trainers on a regular basis (in-service training–professional development).

- Request the input of an architect familiar with accommodations necessary to encourage participation of person's with physical disabilities during the initial planning for any new construction that is funded as part of the WATERS Network, such as monitoring sites,

laboratory space, classrooms, etc. or the renovation of existing facilities. Often these accommodations are straightforward and easily accomplished, such as making doors and space between lab benches wider, lowering benches to a reasonable height for those in a wheelchair and leaving space underneath some benches open to allow wheel and legroom.

9.3 Possible Funding Sources for Education and Outreach

WATERS Network education directors will look for opportunities for collaboration on education initiatives. Many federal governmental education programs are directly applicable including the following.

- NSF Graduate Fellows in K-12 Education, Research Experiences for Undergraduates, Research Experiences for Teachers, and the Integrative Graduate Education and Research Traineeship Program.

- USEPA Environmental Education Grants, Educator Training Program, and the National Network for Environmental Management Studies.

- USDA Education Challenge Programs

The above list is not exhaustive and does not include the numerous opportunities available at the state and local level.

9.4 Possible Organizations for Partnerships

Other Environmental Observatory Systems

- CUASHI
- IOOS
- LTER
- NEON
- ORION

Universities

- HBCUs
- Tribal Colleges
- HSIs
- Researchers
- Centers for Engineering Education Research
- Extension offices

After School Programs

- Girls Inc
- Boys Club
- Girls Club
- Scouts
- 4-H

Informal Education Organizations

- Museums

Community Organizations

NGOs

Federal and State Agencies

- CDCEPA
- NASA
- NIEHS
- NSF
- NOAA
- USGS

Professional Societies

- ASEE
- AEESP
- AAEE
- NSTA

Digital Libraries/Online Partners

- GLOBE - http://www.globe.gov/
- Clubhouse Network - http://www.intel.com/education/icc/
- NSDL The National Science Digital Library - http://nsdlib.org/
- MERLOT: Multimedia Educational Resource for Learning and Online Teaching http://www.merlot.org/
- TeachEngineering http://www.teachengineering.com/index.php

9.5 Organizations from Which to Solicit Input On Education Plan

- AEESP
- NSTA
- ASEE: K-12 Division
- ASEE: Environmental Engineering Division
- ASEE: Educational Research and Methods Division
- IEEE: Education Society
- IEEE: Computer Science
- AGU: Hydrology and Biogeosciences
- ASCE: Environmental Engineering
- AICHE: Environmental Engineering
- ASBAE
- DNR: States
- CAEE
- Union of Concerned Scientists
- Engineers for a Sustainable World

9.6 WATERS Network Education Committee

Beth Eschenbach, Humboldt State University, co- chair
James Johnson, Howard University, co- chair

Chris Brus, University of Iowa
Patricia Carlson, Rose-Hulman Institute of Technology
Daniel Giammar, Washington University
Bette Grauer, McPherson High School
Liesl Hotaling, Stevens Institute of Technology
Gbekeloluwa Oguntimein, Morgan State University
Steven Safferman, Michigan State University
Emma Seiler, Mississippi State University
Tim Wentling, University of Illinois at Urbana-Champaign